

04 — Plastic as a Carbon Storage Medium

How the German lignite opencast mines may become Europe's grandest multi-century carbon storage facilities.

What this instalment covers

- Plastic stability in anaerobic geological environments: proven for 500–5,000 years, running parallel to natural fossil strata.
- The German opencast mines of Garzweiler, Hambach, Inden, and Welzow-Süd: their dimensions, closure dates (2030), volumetric capacities, and geological suitability.
- The circularity that must not be named circular: extracting carbon from lignite, depositing plastic in its stead — the same geological space, an inverted direction of flow.
- Logistical architecture: source separation in urban areas, transport by rail, and sorting and packaging hubs at the perimeter of the opencast mines.
- Costs per tonne of stored plastic versus savings versus incineration: approximately 2.86 tonnes of CO₂ equivalent avoided per tonne, ensuring centuries of sequestration.
- The constitutional and environmental legal reconsideration required for the reclassification of opencast burial as a 'carbon storage facility' rather than a 'landfill'.

This instalment is being further developed with cross-sections of the opencast mines, a logistical map of transport corridors, and the legal arguments for reclassification.